STAT 437 Project: Do Turf Fields Cause an Increase Injuries in Football?

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Introduction
It is a commonly held belief amongst NFL players and fans alike that the use of artificial turf fields in place of natural grass leads to higher rates of serious lower body injuries. However, some dismiss these claims as being a mere myth that is not supported by data. In an internal review done by the NFL in 2022 on the matter they found that injury rates are statistically the same on artificial turf vs. natural surfaces even claiming that "looking at synthetic versus natural doesn't really provide us the information we need to try to drive injury rates down" (Seifert). On the other hand, in a systematic review published in The American Journal of Sports Medicine it was found that while across sports the injury rates for both types of surfaces were similar, for specifically American football players at high levels of competition there was more risk of knee injuries when playing on artificial surfaces (Gould et al). In 2020, the president of the NFL Players Association published an article advocating "for teams to convert artificial practice and game fields to natural grass fields" due to turf fields being "significantly harder on the body than grass" (Tretter). This issue is of pressing concern for NFL players as currently 18 of 32 teams in the league play their home games on artificial turf fields. The aim of this project is to answer the question of whether type of playing surface impacts the rate and severity of player injuries in the NFL.

Data Definitions
In order to answer the causal question we needed data on plays in which players were injured, looking at the type of surface the injury occurred on and the severity of said injury while also having data for potential confounders such as weather. The response variable being used for the data is injuries that sidelined players for 28 days or more (DM_M28 in the data) as such injuries would certainly be considered serious. The explanatory variable is surface type (natural of synthetic). Each row represents a description of the conditions surrounding one specific play in which a player was injured.

Data for Analysis The data used in this analysis is from a Kaggle competition called "NFL 1st and Future - Analytics." It consists of two datasets: Injury Record and Play List. These datasets provide information about player injuries, game details, and play types in the NFL.
Injury Record (ir): This dataset contains information about the injuries suffered by NFL players during the 2016–2018 seasons. Each row in the dataset represents an injury incident, and the columns include details about the player, injury, and game environment. Play List (pl): This dataset contains information about every play in which an injury occurred during the 2016-2018 NFL seasons. It includes details about the player, game, and play type.
To create the dataset ("nf") for the analysis, the following steps were performed:
1. Read the Injury Record and Play List datasets into R data frames (ir and pl, respectively).
2. Drop duplicate rows in the Play List dataset (pl) based on the combination of PlayerKey and GameID, keeping all other columns in the dataset.
3. Merge the Injury Record and Play List datasets on the PlayerKey and GameID columns, keeping all rows from the Injury Record dataset. 4. Drop unnecessary columns from the merged dataset (jd).
5. Calculate the average temperature, ignoring the -999 values, and replace -999 values in the Temperature column with the calculated average temperature.
6. Round the Temperature values by extracting the first two characters of the column.
7. Standardize the names of indoor and outdoor stadiums in the StadiumType column.

8. Standardize the names of different play types in the PlayType column.
9. Drop rows with blank data for indoor or outdoor stadium.
10. Assign NA to Weather and Temperature columns for indoor stadiums.
11. Convert the Temperature column to an integer data type.
12. Remove an unnecessary column named ‘r’.
13. The resulting dataset, nf, is now ready for analysis.

The nf dataset is a cleaned and preprocessed version of the original data, with a focus on the relevant variables for the analysis. It contains information about player injuries, game details, and play types in a consistent and structured format.

Works Cited

Gould, Heath P, et al. “Lower Extremity Injury Rates on Artificial Turf versus Natural Grass Playing Surfaces: A Systematic Review.” The American Journal of Sports Medicine, U.S. National Library of Medicine, 20 May 2022, https://pubmed.ncbi.nlm.nih.gov/35593739/.

Tretter, JC. “Only Natural Grass Can Level the NFL’s Playing Field.” NFL Players Association, 20 Sept. 2020, https://nflpa.com/posts/only-natural-grass-can-level-the-nfls-playing-field. Seifert, Kevin. “NFL Data Shows Recent Injury Rates Same on Grass, Artificial Turf.” ESPN, ESPN, 9 Nov. 2022, https://www.espn.com/nfl/story/_/id/34982032/nfl-data-shows-recent-injury-rates-same-g rass-artificial-turf.

“NFL 1st and Future - Analytics.” Kaggle, The National Football League, 25 Nov. 2019, https://www.kaggle .com/competitions/nfl-playing-surface-analytics/data.